Clustering (HW-01)

K-means clustering is a very famous and powerful unsupervised machine learning algorithm. *A K-means clustering algorithm tries to group similar items in the form of clusters. The number of groups is represented by K.*

The working of K-mean clustering was explained during the lecture in detail. Nevertheless, a high-level K-means algorithm is listed below:

Algorithm 1 k-means algorithm

```
Specify the number k of clusters to assign.
Randomly initialize k centroids.
repeat
    expectation: Assign each point to its closest centroid.
    maximization: Compute the new centroid (mean) of each cluster.
until The centroid positions do not change.
```

Your job is to implement the K-means algorithm in python.

The program should:

- Generate Two-dimensional coordinates (as data points)
    - Either allow a user to enter It Manually OR
    - Automatically generate a few data points
- Allow a user to enter the number of clusters, i.e., the value of K
- Provide a visualization of k-means for each iteration until the Algorithm converges, i.e., optimal clusters are formed

The choice of *Distance Measures* is a critical step in clustering. It defines how the similarity of two elements (x, y) is calculated and will influence the clusters' shape. Please ONLY use Euclidean Distance for the assignment.

CONSTRAINTS:

- You can use **only two** <u>non-Primitive Built-in structures</u> (a) **List** and (b) **Sets**.
- Please DON'T USE any user-defined structures such as Stacks, Queue and tress along with data analysis libraries such as Pandas and Numpy and Pytorch AND Packages Scikits.learn.cluster.KMeans. and Pyclustering.
- You can use ANY graphical Python Library for Visualization, such as: Matplotlib, Seaborn and Plotly

Note:

Assignment Marks : 10 Points

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Over the years, I have compiled a list of many implementations of simple K-means over the internet. Despite my claim, I find many students using the available solution either as it is OR with a slight modification to variables and function names and loops.

Please don't paste from

(a). **Chegg** : https://www.chegg.com/homework-help/questions-and-answers/3-homework-problem-3-implement-k-means-use-methods-point-cluster-implement-missing-kmeans--q74884494 OR
(b). **Kaggle**: https://www.kaggle.com/code/prashant111/k-means-clustering-with-python/notebook (many others at Kaggle) OR
(c). **Stackoverflow**: https://stackoverflow.com/questions/33458834/k-means-clustering-in-python

There is **NO** harm in using them as a **good resource** to understand K-means BUT not for copying.

The homework is to make you implement K-means so that all the concept becomes clear to you. If you copy or cheat, YOU lose, NOT me. I already know clustering 😊

If you don't know where to start or what to discuss your implementation idea with me, DON'T hesitate to ask.

Let your program OUTPUT some Meta-data (possibly for every iteration) that can help someone comprehends the "Happening" for that iteration.

For the boot-strapping purpose, sketch out a few data points and seed a preplanned centroid. So that you can validate the working on the Algorithm. I would suggest working out a mechanism that should converge on the first attempt, just for the sake of validation them stretch it.

Once you validate your code against a preplanned/seeded example of yours, then you can go crazy in adding a large number of points and experiment with different values of K 😊

Deliverables:

Please upload your FULLY COMMENTED CODE, i.e., .py file

A few tests run(s) result as PDF. Meaning, starting data point(s) and their corresponding clustering output.

Assignment Marks : 10 Points